

Axel Refalo

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Robotics and research enthusiast, holding a Master's in Automation Engineering (ÉTS) and a Master's in Mechanical Engineering (UTBM). Seeking a position in a robotics-related startup

EDUCATION

École de Technologie Supérieure – ÉTS

M.Sc. in Automation Engineering, research-based

Montréal, Canada

Sept. 2024 – now

- Master's thesis on kinematic singularities and path planning for serial robot arms at CoRo lab
- Publication at the IEEE International Conference on Robotics and Automation
- 5th Summer School on Singularities of Mechanisms and Robotic Manipulators

Vienna – 2026

Barcelona – 2025

McGill University

Computer Science, exchange program

Montréal, Canada

Sept. 2024 – May 2025

- Graduate courses in English
- Applied Machine Learning, applied projects for pattern recognition and classification in Python involving tools like Linear and Logistic Regression, K-Nearest Neighbors, Decision Trees, Convolutional Neural Networks, Deep Neural Networks, and Large Language Models
- Applied Robotics, Implemented controller packages on Robot Operating System with Gazebo in C++ for dynamic and kinematic control of the redundant robotic arm, the Kinova Gen3.

University of Technology of Belfort-Montbéliard – UTBM

B.Eng. and M.Eng. in Mechanical Engineering, specialization in Mechatronics

Belfort, France

Sept. 2020 – Feb. 2026

- Science courses covering mechanics (control systems, continuum mechanics, finite element methods, computer-aided design and additive manufacturing), electronics (electronic design, signal processing) and computer science (LabVIEW, C++ and C# languages and calculus)
- Supplemented by humanities: English, Spanish, Sustainable Development and History

University of Palermo

Mechanical Engineering, exchange program

Palermo, Italy

Sept. 2021 – Feb. 2022

- Undergraduate courses in Italian on kinematics and dynamics of mechanisms

RESEARCH

Control and Robotics Laboratory, CoRo – ÉTS

Master's thesis

Montréal, Canada

May 2025 – now

- Comparative analysis of singularities, manipulability and cuspidality for six-revolute-joint collaborative robots from Universal Robot, Doosan, ABB and Fanuc manufacturers
- Implemented an Inverse Kinematics solver for a general six-revolute-joint serial manipulator which interfaces with RoboDK (C++)
- Publication: Refalo, A., Bonev, I., and Gosselin, C. "Singularity Analysis of ABB's GoFa 5 Robot Arm," accepted at International Conference on Robotics and Automation 2026

Laboratory of Bioresources – McGill

Research project

Montréal, Canada

Jan. 2025 – now

- Robotic platform for collection of plant growth data (camera and sensors) and to model spatiotemporal growth patterns, design and control from scratch
- In preparation: Debbagh, M., Refalo, A., et al. "An Affordable and Open-Source Gantry Framework for Capturing High Resolution Spatiotemporal Plant Growth Data"

Materials Science Laboratory – ÉTS

Research project

Montréal, Canada

Dec. 2024 – May 2025

- Crack propagation analysis for different concrete mixes, focusing on computed tomography reconstruction of synchrotron concrete scans and filtering using a Convolutional Neural Network

- Kalman filter derivation and implementation for position estimation using signals of opportunity from Low Earth Orbit satellite constellations

- Microscopic and high-speed ink behavior investigation on nozzles of large-format printers
- Two Cartesian robots development: (I) for inspecting static microscopic print nozzle defects and contamination, and (II) for inspecting dynamic behavior of the ink on the coating surface
- Research fields: image filtering, spectral analysis, electronic and mechanical design of precise robotic axes and control interface with LabVIEW

PROJECTS

Robotics

- Two-cable-driven parallel plotter for spray painting: dynamic model derived analytically, validated and implemented 2026 2024
- Three-degree-of-freedom manipulator with parallelogram mechanism, for pick-and-place tasks, forward and inverse kinematics derived analytically and implemented 2022
- Wide planar plotter driven differentially with one belt to draw with a pen
- Other robotics projects: simple robot arm, modular robot chain, hovercrafts, and electric skateboards

Computer Aided Design

- Mechanical and electronic design and build of all previous robotics projects
- Replication of a Swiss watch mechanism, design of parametric gears with a cycloidal profile 2023

Programming

- All the previous robotics projects, coded mostly in C++ or Python
- Home lab server built and accessed remotely (Unix Shell) 2025
- Simple Eulerian fluid simulator (C++) 2023
- Algorithms to solve complex grid puzzles: sudoku, kakuro, killer sudoku, suguru using backtracking (C++) 2023
- Mechanical truss simulator using finite element methods (C#) 2022

SKILLS

Languages	French (Native), English (Bilingual - IELTS), Italian (Bilingual - IB), Spanish (Beginner)
Software	Robotics: Robot Operating System, Gazebo, RoboDK, RobotStudio and Roboguide Programming: Python, C++, C#, LabVIEW, Unix shell, LaTeX and Visual Studio Code Math: MATLAB, SciPy, Wolfram, Maple and GeoGebra
Mechanics	Computer Aided Design: CATIA V5, Fusion 360 and AutoCAD Physical modeling: Simulink, Amesim, Ansys, OrCAD, Abaqus and DragonFly
Electronics	Controller board (Raspberry Pis and Arduinos) motors (stepper and brushless), batteries and power converters, and usual electronic components
Manufacturing	Additive printing, laser cutter, electronic soldering and hand tools

INTERESTS

Driven by challenge and curiosity · **Travel** across Canada, Netherlands, Italy and Spain · **Arts**: literature, cinema, documentary and drawing · **Sport**: chess, hiking and swimming · And a deep passion for applying **science** through collaborative engineering challenges, including:

- Employed as FabLab manager, design department of ÉTS 2025
- CrunchTime hackathon hosted by UTBM: participant (2023) and electronics manager (2024)
- Crunch industry camp hackathon hosted by UTBM: finalist 2022